Started on	Wednesday, 16 October 2024, 9:22 PM
State	Finished
Completed on	Sunday, 27 October 2024, 4:55 PM
Time taken	10 days 19 hours
Marks	10.00/10.00
Grade	**100.00** out of 100.00
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Question 1

Correct

Mark 10.00 out of 10.00

Write a java program which performs the following.

- Create a super class called Car. The Car class has the following fields and methods.
  - int speed;
  - double regularPrice;
  - String color;
  - double getSalePrice();
- Create a sub class of Car class and name it as Truck. The Truck class has the following fields and methods.
  - int weight;
  - double getSalePrice(); //**If weight>2000,10%discount. Otherwise, 20%discount.**
- Create a subclass of Car class and name it as Hatchback. The Ford class has the following fields and methods.
  - int year;
  - int manufacturerDiscount;
  - double getSalePrice();**//From the saleprice computed from Car class, subtract the manufacturer Discount.**
- Create a subclass of Car class and name it as Sedan. The Sedan class has the following fields and methods.
  - int length;
  - double getSalePrice(); **//If length>20 feet, 5% discount, Otherwise,10%discount.**
- Create MyOwnAutoShop class which contains the main() method. Perform the following within the main() method.
  - Create an instance of Sedan class and initialize all the fields with appropriate values.
  - Create an instance of the Hatchback class and initialize all the fields with appropriate values.
  - Create an instance of the Truck class and initialize all the fields with appropriate values.
  - Create an instance of Car class and initialize all the fields with appropriate values and Display the sale prices of all instance.

For example:

Input	Result
155 85000 Green 2200 180 120000 White 7 175 75000 Grey 1998 5000 155 50000 Yellow	Enter Speed,regularPrice, color, weight for Truck Truck Price 76500.00 Enter Speed,regularPrice, color, length for Sedan Sedan Price 108000.00 Enter Speed,regularPrice, color, year, manufacturerDiscount for Hatchback Hatchback Price 70000.00 Enter Speed,regularPrice, color for Car Basic Car Price 50000.00

Answer: (penalty regime: 0 %)

```
import java.util.Scanner;
class Car {
    int speed;
    double regularPrice;
    String color;

    double getSalePrice() {
        return regularPrice;
    }
}

class Truck extends Car {
    int weight;
    double getSalePrice() {
        if (weight > 2000) {
            return regularPrice * 0.90;
        } else {
            return regularPrice * 0.80;
        }
    }
```

```
}

class Hatchback extends Car {
    int year;
    int manufacturerDiscount;
    double getSalePrice() {
        return regularPrice - manufacturerDiscount;
    }
}

class Sedan extends Car {
    int length;
    double getSalePrice() {
        if (length > 20) {
            return regularPrice * 0.95;
        } else {
            return regularPrice * 0.90;
        }
    }
}

public class MyOwnAutoShop {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.println("Enter Speed,regularPrice, color, weight for Truck");
        Truck truck = new Truck();
        truck.speed = scanner.nextInt();
        truck.regularPrice = scanner.nextDouble();
        truck.color = scanner.next();
        truck.weight = scanner.nextInt();
        System.out.printf("Truck Price %.2f%n", truck.getSalePrice());

        System.out.println("Enter Speed,regularPrice, color, length for Sedan");
        Sedan sedan = new Sedan();
        sedan.speed = scanner.nextInt();
        sedan.regularPrice = scanner.nextDouble();
        sedan.color = scanner.next();
        sedan.length = scanner.nextInt();
        System.out.printf("Sedan Price %.2f%n", sedan.getSalePrice());

        System.out.println("Enter Speed,regularPrice, color, year, manufacturerD
        Hatchback hatchback = new Hatchback();
        hatchback.speed = scanner.nextInt();
        hatchback.regularPrice = scanner.nextDouble();
        hatchback.color = scanner.next();
        hatchback.year = scanner.nextInt();
        hatchback.manufacturerDiscount = scanner.nextInt();
        System.out.printf("Hatchback Price %.2f%n", hatchback.getSalePrice());

        System.out.println("Enter Speed,regularPrice, color for Car");
        Car car = new Car();
        car.speed = scanner.nextInt();
        car.regularPrice = scanner.nextDouble();
        car.color = scanner.next();
        System.out.printf("Basic Car Price %.2f%n", car.getSalePrice());

        scanner.close();
    }
}
```

	Input	Expected	Got	
✔	155 85000 Green 2200 180 120000 White 7 175 75000 Grey 1998 5000 155 50000 Yellow	Enter Speed,regularPrice, color, weight for Truck Truck Price 76500.00 Enter Speed,regularPrice, color, length for Sedan Sedan Price 108000.00 Enter Speed,regularPrice, color, year, manufacturerDiscount for Hatchback Hatchback Price 70000.00 Enter Speed,regularPrice, color for Car Basic Car Price 50000.00	Enter Speed,regularPrice, color, weight for Truck Truck Price 76500.00 Enter Speed,regularPrice, color, length for Sedan Sedan Price 108000.00 Enter Speed,regularPrice, color, year, manufacturerDiscount for Hatchback Hatchback Price 70000.00 Enter Speed,regularPrice, color for Car Basic Car Price 50000.00	✔

Passed all tests! ✔

▸ **Show/hide question author's solution (Java)**

Correct

Marks for this submission: 10.00/10.00.